
accelerate
collaborate
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palm
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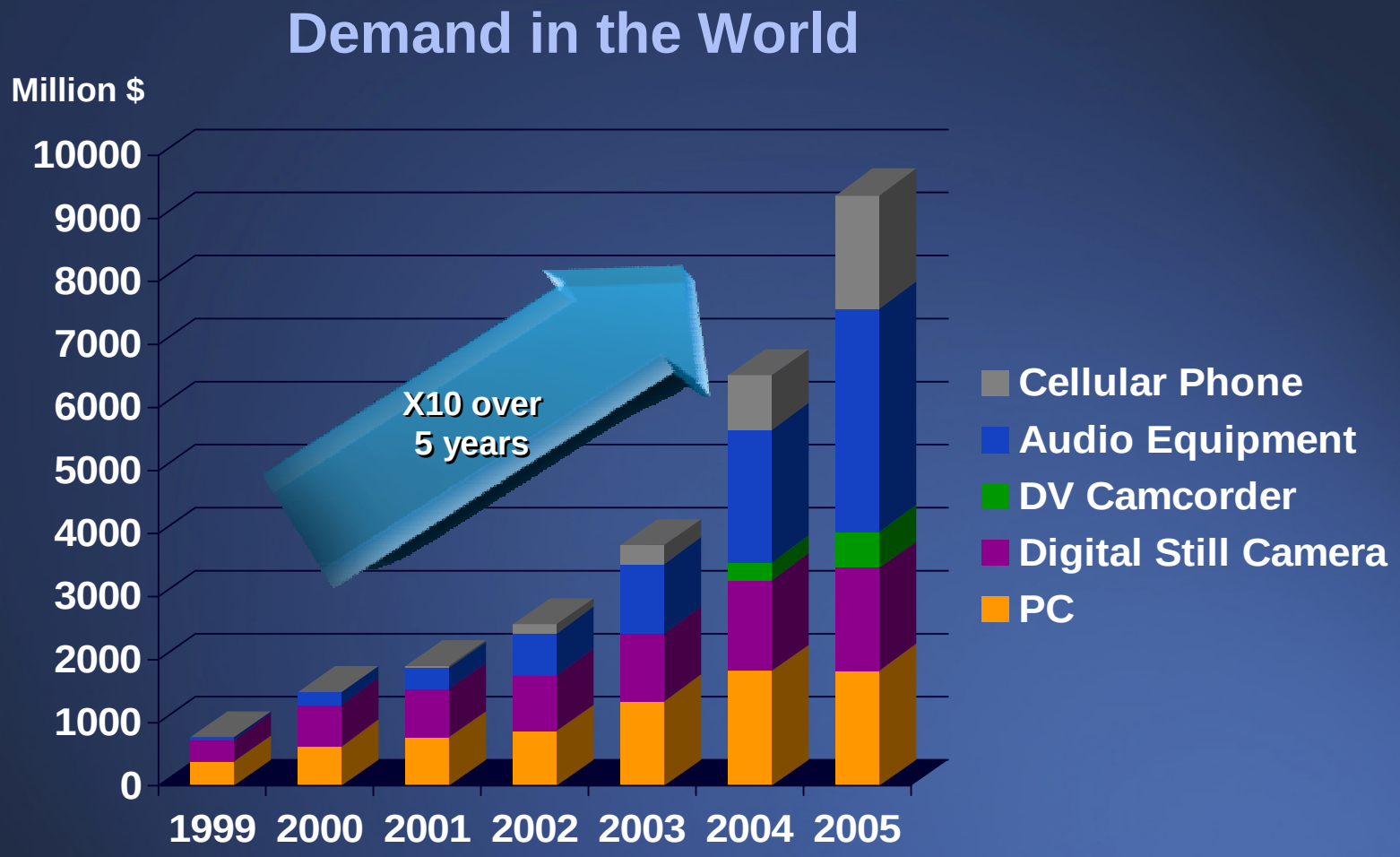


Palm OS® Support for Expansion and Secondary Storage—Part II

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Palm Developer Support

PalmSource
Conference 2000

World Demand for Memory Cards



Source: Technology System Research

Interoperability of Memory Cards



Audio & Visual

TV/DVC/DSC
Set Top Box
Portable Audio Equipment
DVD Audio Equipment

Mobile

Cellular/PHS Phone/PDA
Next Generation Mobile
Communication Terminal
Car Navigation System
Car Audio Equipment





Compact

PC

Personal Computer Printer
PC Card Adapter
Scanner

Large Capacity



Secure

Telecommunications

Internet Phone
Cordless Phone
Fax/E-Mail Terminal

High Speed



Entertainment Game, etc.

KIOSK Terminal
Health Care &
Medical Equipment

Home Appliance

Microwave Oven
Rice Cooker/Refrigerator
Air Conditioner
Air Purifier







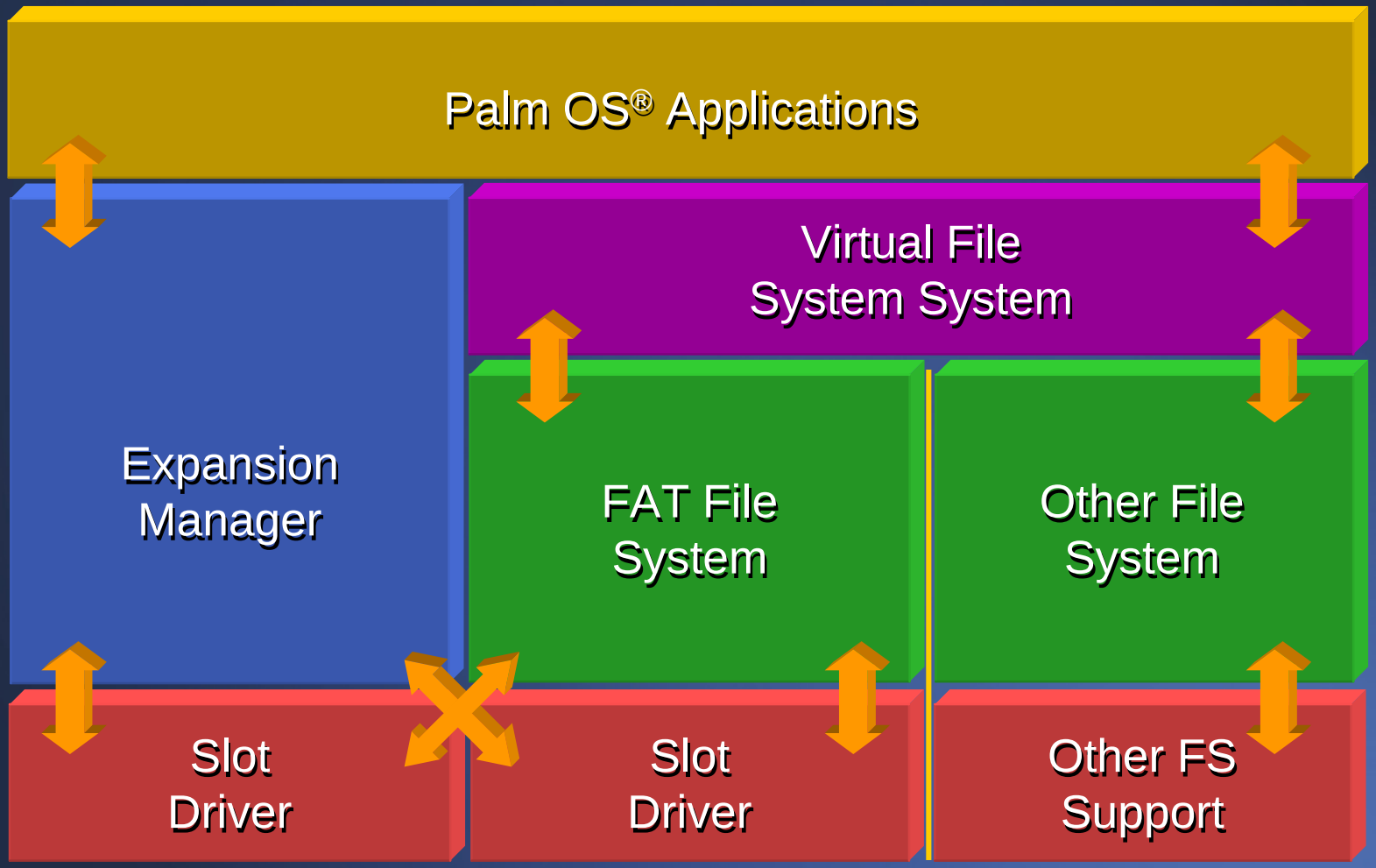
Key Palm OS® Features

- Hot insertion and removal of cards/devices
- Support for various expansion media
  - MultiMediaCard, SecureDigital, CompactFlash, MemoryStick, etc.
- Enable “auto-launch” application
- FAT12/16 file system support

Key Palm OS® Features

- Supports multiple cards in one device
- Extends HotSync® API to provide access to expansion filesystem
- Integrated with Palm OS® Launcher

Functional Block Diagram



Preparing an App for a Card

- Card-related Launch Codes
  - Some applications want to know when they are launched from card
- Bundling Databases on Launch
  - Some applications include bundled databases
  - Especially important for overlays
- Auto-Launch of an Application
- Other Considerations

Detecting Launch From Card

- The sysAppLaunchCmdCardLaunch launch code is sent to applications after being launched from card
- If the sysAppLaunchCmdCardLaunch launch code is not handled by an application, then a sysAppLaunchCmdNormalLaunch is sent

Detecting Launch From Card

```
case sysAppLaunchCmdNormalLaunch:
   error = AppStart();
   if (error)
      return error;

   FrmGotoForm(MainForm);
   AppEventLoop();
   AppStop();

   break;
```

Detecting Launch From Card

```
case sysAppLaunchCmdCardLaunch:
   pParms = (SysAppLaunchCmdCardType*)cmdPBP;
   pParms->err = errNone;

case sysAppLaunchCmdNormalLaunch:
   error = AppStart();
   if (error)
      return error;

   FrmGotoForm(MainForm);
   AppEventLoop();
   AppStop();

   break;
```

Bundling Databases on Launch

- An application can have “bundled” databases, which are copied, deleted and beamed with it
- A database is bundled if all of the following
  - It has the same creator id as the application
  - Does not have dmHdrAttrCopyPrevention set
  - Either has dmHdrAttrBundle set or is of type sysFileTOverlay
- Bundled databases are copied to main memory on launch from card

Auto-Launch of an Application

- An application can be auto-launched when card is inserted
  - Auto-start application is named /Palm/start.prc
  - VFS Manager registers for the “volume mounted” notification at lower than default priority
  - Looks for start.prc and launches it

Preventing Auto-Launch

Handle the “volume mounted” notification

Some apps might want to use the card like a floppy, for data storage, so quitting on card insertion or removal wouldn’t be ideal

Preventing Auto-Launch

Handle the “volume mounted” notification

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```
PrvVolMountedNotifyProc(SysNotifyParamType
  *notifyParmsP)
{
  notifyParmsP->handled |= vfsHandledUIAppSwitch;
  return(0);
}
```

Security During Auto-Launch

- If device is locked when auto-launch application is launched, user is prompted for security password prior to launch of start.prc

Other Considerations

- Applications on secondary storage do not receive launch codes or notifications
- A good example is “Find” functionality
- To receive and handle launch codes, apps should install a “stub” application on the device to register and receive these launch codes or notifications

Working With Data on Cards

- Directory Mapper
  - Registry for mapping common file types (mime types) to specific directories on cards
- Palm OS® Database functions
  - VFSImportDatabaseFromFile
  - VFSExportDatabaseToFile
- Consider using Filesystem APIs for conventional file access or apps with very large data requirements

Directory Mapper

- Registry for mapping common file types (mime types) to specific directories on cards
- Different media have a different location for data
  - Images (.jpg) is in “/DCIM” on Compact Flash
- Launcher registers “.prc”, “.pdb”, “.pqa” file types for all media types
- Avoid forcing the user to navigate the card by full pathname

PRC-Only Application on Card



PRC Application





- Palm OS 4.0:
  - Displays application in Launcher
  - Launch application from card
- Application Modifications:
  - None

PRC-Only Application on Card

- Application (PRC) is copied into main memory on launch
- A “CardLaunch” launch code is sent. If not handled, a “NormalLaunch” launch code is sent
- PDB and/or preferences may be created in main memory
- Application completes and is deleted
- PDB and/or Preferences remain in main memory

PRC+PDB Application on Card



Launcher App PRC And PDB





- Palm OS 4.0:
  - Displays app in Palm OS® Launcher
  - Launch application from card
- Application Modifications:
  - Minimal Level of Effort
    - Developer sets “Bundled” bit on all PDBs on card
    - PRC and PDB(s) are copied to main memory

PRC+PDB Application on Card

- Application (PRC) and all bundled databases (PDB's or overlays) are copied into main memory on launch
- A "CardLaunch" launch code is sent. If not handled, a "NormalLaunch" launch code is sent
- Application completes and is deleted
- PDB and/or Preferences remain in main memory

Launcher Demo



App With Large Data Requirements

```
Use Standard VFS File I/O

// Reading file data
VFSFileReadData(FileRef, numBytes, &buf,
                  offset, &numBytesRead)
```

```
Use Data Manager for PDB resident in Memory

// Get the current record.
memoRec = DmGetRecord (MemoDB, CurrentRecord);
```

```
Use VFS Utilities for PDB resident on card

// Get the current record.
VFSFileDBGetRecord(FileRef, CurrentRecord, &recH,
          &recAttr, &uniqueID)
```

- Palm OS 4.0:
  - Displays app in Launcher
  - Launch application from card
- Application Modifications:
  - Moderate Level of Effort
    - Data is read from card via File I/O (or PDB File I/O)
    - Index management of content on card

App Using Alarms or Launch Codes

Handling the Alarm

```
Avoid launching app from card.
Store data for retrieval later.
```

Hiding the stub application

```
DmDatabaseInfo(…, &attribs, …);
attribs |= dmHdrAttrHidden;
DmSetDatabaseInfo(…, attribs, …);
```

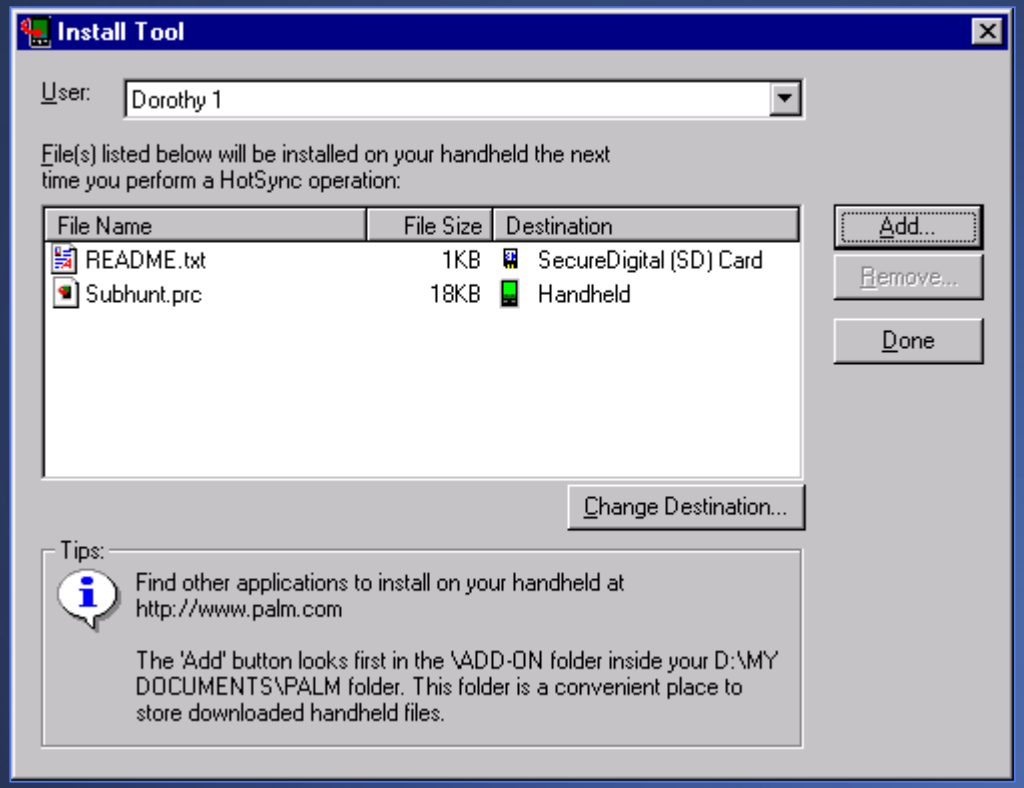
- Palm OS 4.0:
  - Displays app in Launcher
  - Launch application from card
- Application Modifications:
  - Moderate level of effort
    - Create a small stub application to handle alarms or launch codes

HotSync® Manager Enhancements

- Extended to support secondary storage
- New Sync Manager APIs
  - VFS Manager APIs
  - Expansion Manager APIs
- New Install Aide and UserData APIs
  - File installation to and from device

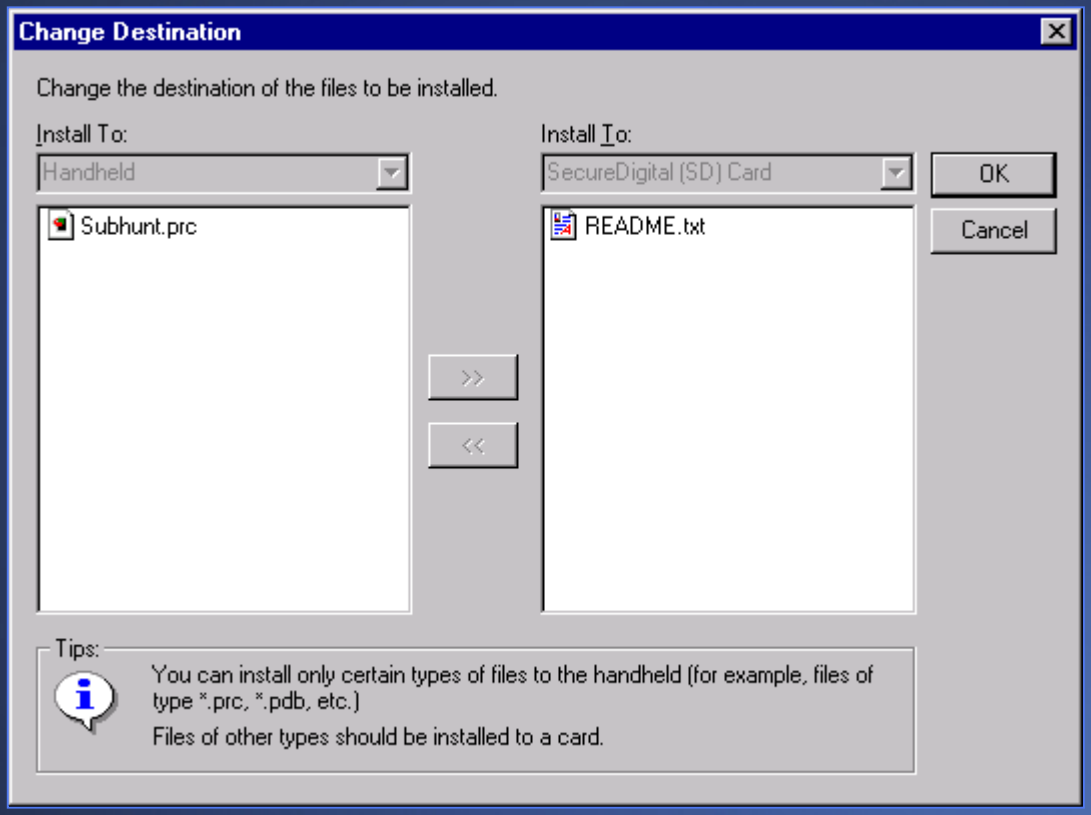
Palm Install Tool Enhancements

- Modified UI to show expansion



Palm Install Tool Enhancements

- Modified UI to show expansion



Palm Install Tool Enhancements

- Support for “non-Palm” filetypes (.jpeg, .mp3, etc.)
  - Palm Filetypes (“.prc”, “.pqa”, “.pdb”) default to “/Palm/Launcher” directory if they are installed to card
  - Non-Palm filetypes default to location registered in Directory Mapper, or fail to get installed if filetype not registered
- New install conduit with expansion support

POSE® Support

- VFS Applications can be developed on POSE
- RamDisk
  - Creates a card in storage heap
- HostFS
  - Uses native filesystem as a card (PC or Mac)
  - PRC available for download
    - HostFS.prc
    - File system implementation

Summary

- Palm OS® will include expandable architecture for support of various expansion technologies
- Consistent User Experience
- Keep it simple!
  - But… we need your help!
- Interoperability is key!


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